

Generation Interconnection System Impact Study Report

For Queue Project Q21-1879

Filbert 138 kV

June 25, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1879. The Interconnection Customer proposes a 158.3 MW solar photovoltaic facility interconnecting at the Filbert 138 kV substation in the New Jersey area. The request was received on September 08, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 497 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$5,512,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Mccree / Coks Corner 138kV
Contingency: Mathewson / Mount Laurel 345kV Line
Loading: 111.1% (Rating: 203 MVA)
DFAX: 31.5%
Pre-project loading: 86.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mccree / Coks Corner 138kV	Rebuild line with higher-capacity conductor	\$8,996,000	61% (\$5,512,000)
	Total allocated to project		\$5,512,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.